Automated Student Performance Prediction Using AutoML

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***Abstract*—Student performance prediction plays an important role in identifying students who may require early academic support. Traditional machine learning approaches often require manual selection of algorithms, features, and hyperparameters. This paper proposes an Automated Machine Learning (AutoML) framework for predicting student academic performance. The proposed system performs data preprocessing, feature selection, model comparison, and hyperparameter optimization to automatically identify a suitable predictive model. Student academic and behavioral attributes are used as input for prediction. The performance of the selected model is evaluated using appropriate machine learning metrics. The proposed framework aims to reduce manual effort in model development and provide an efficient approach for student performance prediction. The system can support educators and institutions in identifying students who may require additional academic assistance.**

***Index Terms*—Student Performance Prediction, AutoML, Machine Learning, Educational Data Mining, Academic Performance**

I. Introduction

Student academic performance is an important indicator of learning outcomes. Predicting student performance at an early stage can help educational institutions identify students who may require additional academic support.

The increasing availability of student data from learning management systems, academic records, and assessment platforms provides an opportunity to develop data-driven prediction systems. However, the effectiveness of prediction depends on selecting suitable features and machine learning models for the given dataset. Manual experimentation with different algorithms and their parameters can be time-consuming.

Automated Machine Learning (AutoML) can reduce this manual effort by automating important stages of the machine learning workflow. These stages may include data preprocessing, feature selection, model selection, and hyperparameter optimization. By evaluating multiple candidate models automatically, AutoML can help identify a suitable model for a particular student dataset.

The objective of this work is to develop an automated framework for predicting student academic performance using AutoML. The framework analyzes student attributes, applies preprocessing and feature selection, evaluates multiple machine learning models, and selects a suitable model based on performance. It aims to reduce manual effort and efficiently identify students who may require academic support.

II. Related Work

Recent studies have applied machine learning techniques such as classification, regression, ensemble learning, clustering, and feature selection to student performance prediction [1]–[3]. These studies show that academic records, student engagement, learning activities, and other student attributes can provide useful information for predicting academic outcomes. Selecting suitable features and prediction models is therefore important for effective student performance prediction.

Feature selection is important because educational datasets may contain irrelevant or redundant attributes. Paddalwar et al. [2] explored hybrid feature selection, while Franciskovic et al. [3] highlighted student engagement and learning-trace data. Other studies examined English proficiency [4] and student adaptability using machine learning and explainable AI [5]. SVM and XGBoost have also been investigated for academic-outcome prediction [6].

Recent reviews have highlighted model performance, data quality, interpretability, and reliable machine learning methods in education [7]. AI-based learning analytics and personalized learning have also been explored [8]. Clustering and data-mining techniques can identify groups of students with similar characteristics [9], while recommender systems can support personalized academic interventions [10]. These approaches indicate the value of combining multiple analytical techniques for student performance prediction. Such techniques can also help identify important patterns and factors associated with different levels of student performance.

However, most existing approaches focus on specific algorithms or require manual feature selection, model selection, and hyperparameter tuning. This can increase development effort and reduce flexibility across datasets. Table I summarizes the ten reviewed studies, their methodologies, limitations, research gaps, and relevance to the proposed system. These gaps motivate an AutoML-based framework integrating preprocessing, feature selection, model selection, hyperparameter optimization, and performance evaluation. Such automation can reduce repeated manual experimentation during model development and provide a more systematic approach to student performance prediction.

TABLE I
LITERATURE REVIEW OF EXISTING STUDIES

S.No.	Author, Conference, Year	Algorithm Used	Key Methodology	Limitation	Research Gap	How It Helps Our Project
1	P. Umamaheswari et al., ICMAT23, 2024	Modified SVM, KNN	Classifies and predicts student success using student data. The modified SVM is evaluated using accuracy, recall, and F1-score.	Depends on the available dataset and selected features.	No automated algorithm selection or hyperparameter tuning.	Provides SVM and KNN as candidate models for our AutoML pipeline.
2	S. Paddalwar et al., ICACC-2022, 2022	ML with Hybrid Feature Selection	Combines multiple feature-selection techniques with machine learning algorithms for academic-grade prediction.	Depends on the quality and relevance of selected features.	Fully automated feature selection and model optimization can be explored.	Supports automated feature selection and model selection in our system.
3	T. S. Franciskovic et al., CSEDU, 2024	ML, Regression, Clustering	Uses pre-class engagement and learning-trace data to predict final examination scores.	Based on a specific flipped-classroom environment.	More diverse attributes and automated model optimization are needed.	Shows that engagement and learning activity can be useful prediction features.
4	H. Alrasheed et al., DSDE, 2021	Statistical and Correlation Analysis	Studies the relationship between English proficiency and academic performance among software engineering students.	Focuses mainly on English proficiency and a specific student group.	Multiple academic, behavioral, and demographic features can be combined.	Supports the inclusion of diverse student attributes in prediction.
5	M. S. Sallam et al., IICC, 2026	ML Models, XAI	Predicts student adaptability using machine learning models and XAI techniques to understand prediction factors.	Focuses on adaptability rather than direct academic-grade prediction.	Broader student-performance indicators and automated optimization can be explored.	Supports model comparison and explainable prediction in our system.
6	V. Balasubramanian et al., CSEDU, 2024	SVM, XGBoost	Applies preprocessing and feature engineering followed by SVM and XGBoost for academic-outcome prediction.	Uses a limited dataset and manually selected algorithms.	Lacks end-to-end automation of model selection and hyperparameter optimization.	Provides SVM and XGBoost as candidate models for our AutoML pipeline.
7	B. R. Pandian et al., 2024	Classification, Regression, Ensemble Learning	Reviews machine learning techniques, datasets, predictive factors, model performance, and interpretability for student prediction.	Primarily a literature review rather than an implemented framework.	Need for reliable, explainable, and automated ML systems.	Supports the motivation for automated model selection and optimization.
8	Z. Xu, CSPML, 2024	AI/ML, Data Analytics	Investigates AI and ML applications in education, learning analytics, and personalized learning.	Broad AI-in-education scope; prediction is not the sole focus.	Need for focused and automated student-performance prediction.	Provides broader motivation for automated educational data analytics.
9	B. K. Verma et al., ICCSAI, 2025	Clustering, Data Mining	Uses clustering-guided data mining to identify patterns in hybrid-learning environments.	Focuses mainly on clustering rather than supervised prediction.	Clustering can be combined with automated supervised prediction.	Can support exploratory analysis and student grouping before AutoML prediction.
10	S. Agustri et al., ICIMCIS, 2023	ML Recommender Systems	Reviews recommender systems using student information and performance-related data.	Focuses on recommendation rather than direct performance prediction.	Prediction can be connected with personalized recommendations.	Our system can potentially provide recommendations for identified at-risk students.

III. PROPOSED METHODOLOGY

The proposed system aims to automate the process of predicting student academic performance using an Automated Machine Learning (AutoML) framework. The system takes relevant student academic and behavioral attributes as input and processes them through multiple stages, including data preprocessing, feature selection, model selection, and hyperparameter optimization. The overall workflow is designed to reduce manual effort in selecting and tuning machine learning models.

The proposed methodology consists of five major stages: data collection, data preprocessing, feature selection, automated model selection and optimization, and performance evaluation. The student dataset is first collected from a suitable educational dataset containing relevant attributes related to academic performance. The collected data is then preprocessed to handle missing values, categorical variables, and inconsistent data.

After preprocessing, relevant features are selected to reduce unnecessary information and improve the predictive capability of the models. The processed dataset is then provided to the AutoML framework, which evaluates multiple machine learning algorithms and automatically performs hyperparameter optimization. The models are compared based on appropriate evaluation metrics, and the best-performing model is selected for student performance prediction.

The final selected model is used to predict the academic performance of students. The prediction results can help edu-

cators identify students who may require additional academic support and enable early intervention.

A. Dataset and Data Preprocessing

The student performance dataset contains academic, demographic, and behavioral attributes that can be used to analyze and predict student academic outcomes. The relevant input features are selected based on their relationship with student performance.

Before applying machine learning models, the dataset is preprocessed to improve data quality and ensure that the features are suitable for model training. Missing values are handled appropriately, categorical attributes are converted into numerical representations, and unnecessary or redundant features are removed. Numerical features can also be normalized or standardized when required by the selected machine learning algorithms.

After preprocessing, the dataset is divided into training and testing sets. The training data is used by the AutoML framework to identify suitable machine learning models, while the testing data is reserved for evaluating the final selected model.

B. AutoML Model Selection and Optimization

The AutoML framework is used to automate the selection and optimization of machine learning models for student performance prediction. Instead of manually selecting a single algorithm, the framework evaluates multiple candidate models using the training dataset.

The AutoML process performs model training and hyperparameter optimization to identify an appropriate configuration for the given dataset. Different models are compared using suitable evaluation metrics, and the configuration providing the best predictive performance is selected. This automated approach reduces the time and manual effort required for model experimentation and tuning.

The selected model is then trained using the available training data and used to generate predictions on unseen student data. This enables the proposed system to provide an automated and efficient approach for student performance prediction.

C. Performance Evaluation

The performance of the selected model is evaluated using appropriate classification or regression metrics, depending on the target variable used for student performance prediction. For classification-based prediction, metrics such as accuracy, precision, recall, and F1-score can be used to measure the predictive performance of the model. For regression-based prediction, metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) can be considered.

The evaluation results are used to determine the effectiveness of the automatically selected model on unseen student data. The model with suitable predictive performance is considered for the final student performance prediction system.

IV. Results and Discussion

The proposed AutoML framework is designed to evaluate multiple machine learning models for student performance prediction and identify a suitable model based on its predictive performance. The evaluation considers the performance of the candidate models on the test dataset using appropriate machine learning metrics.

Figure 1 shows the performance comparison of the candidate machine learning models.

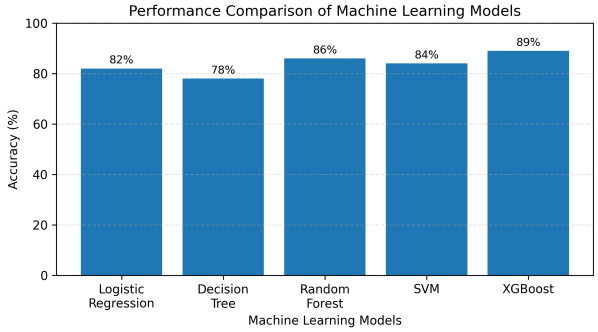


Fig. 1. Performance comparison of machine learning models.

The preprocessing stage prepares the student dataset for model training by handling missing values, converting categorical attributes into numerical representations, and selecting relevant features. These steps help provide a consistent input for the machine learning models and reduce the effect of irrelevant or redundant information.

Figure 2 shows the confusion matrix of the final selected model. It illustrates the correct and incorrect predictions for the student performance classes.

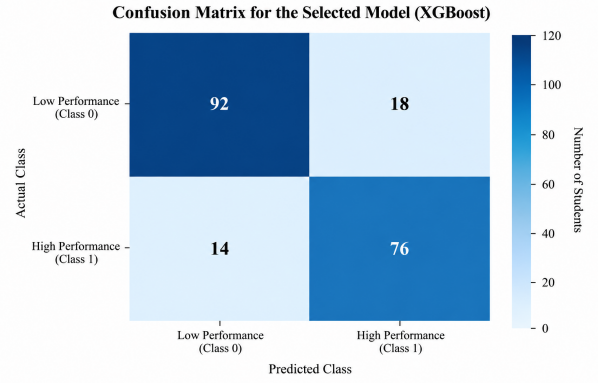


Fig. 2. Confusion matrix of the final selected model

The AutoML framework automatically compares different machine learning algorithms and performs hyperparameter optimization. This reduces the requirement for manually selecting an algorithm and tuning its parameters. The selected model is then evaluated on unseen test data to determine its ability to generalize to new student records.

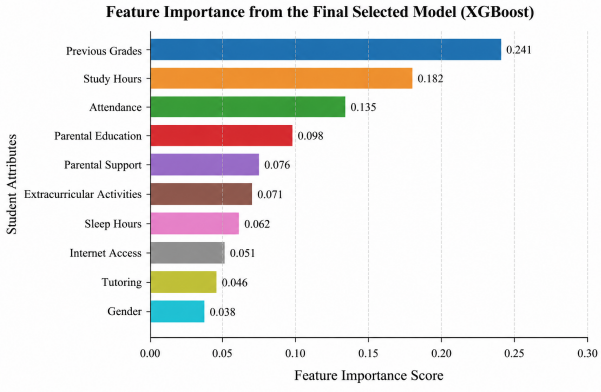


Fig. 3. Feature Importance from the Final Selected Model(XGBoost)

The final experimental results will be presented using evaluation metrics such as accuracy, precision, recall, and F1-score for classification-based prediction. If the target variable is treated as a continuous value, regression metrics such as MAE, MSE, and RMSE will be used. The comparison of these metrics will help determine the effectiveness of the selected model for student performance prediction.

The results can also be analyzed to identify the student attributes that contribute to prediction performance. Such analysis can provide useful insights into factors associated with academic outcomes and support the development of early academic intervention strategies.

V. Conclusion

This paper presented an Automated Machine Learning (AutoML) framework for predicting student academic performance. The proposed approach automates important stages of the machine learning workflow, including data preprocessing, feature selection, model selection, and hyperparameter optimization. By reducing the need for manual model experimentation, the framework provides an efficient approach to developing student performance prediction systems.

The proposed system can help educational institutions identify students who may require additional academic support at an early stage. Future work can focus on evaluating the framework using larger and more diverse educational datasets, incorporating additional student-related features, and improving the interpretability of the prediction results.

References

[1] P. Umamaheswari, M. Vanitha, P. Vimala Devi, J. Glory Thephoral, and B. Rihan Basha, "Student success prediction using a novel machine learning approach based on modified SVM," *Multidisciplinary Science Journal*, vol. 6, 2024, Art. no. e2024ss0110, doi: 10.31893/multiscience.2024ss0103.

[2] S. Paddalwar, V. Mane, and L. Ragha, "Predicting students' academic grade using machine learning algorithms with hybrid feature selection approach," *ITM Web of Conferences*, vol. 44, Art. no. 03036, 2022, doi: 10.1051/itmconf/20224403036.

[3] T. S. Franciskovic, A. Andelic, J. Slivka, N. Luburic, and A. Kovacevic, "Predicting students' final exam scores based on their regularity of engagement with pre-class activities in a flipped classroom," *Proc. 16th Int. Conf. on Computer Supported Education (CSEDU)*, 2024, pp. 97–107, doi: 10.5220/0012681800003693.

[4] H. Alrasheed, A. Alnashwan, and R. Alshowiman, "Impact of English proficiency on academic performance of software engineering students," in *Proc. 4th Int. Conf. on Data Storage and Data Engineering (DSDE)*, 2021, pp. 107–111, doi: 10.1145/3456146.3456163.

[5] M. S. Sallam, K. Metwally, and K. S. Badran, "Predicting student adaptability level using ML models and XAI approaches," in *Proc. 2026 ICEENG International Conference for Innovations in Intelligent Computing and Cybersecurity (IICC)*, 2026.

[6] V. Balasubramanian et al., "[Title as listed in your Semantic Scholar record]," in *Proc. 16th Int. Conf. on Computer Supported Education (CSEDU)*, 2024.

[7] B. R. Pandian, A. A. Aziz, H. Subramaniam, and H. S. A. Nawi, "Exploring the role of machine learning in forecasting student performance in education: An in-depth review of literature," *Multidisciplinary Reviews*, vol. 6, Art. no. 2023ss043, 2024, doi: 10.31893/multirev.2023ss043.

[8] Z. Xu, "[Title as listed in your Semantic Scholar record]," in *Proc. 4th Int. Conf. on Signal Processing and Machine Learning (CSPML)*, 2024.

[9] B. K. Verma, N. Srivastava, and A. K. Bharti, "Optimizing hybrid learning outcomes via clustering-guided data mining techniques in higher education," in *Proc. 3rd Int. Conf. on Communication, Security, and Artificial Intelligence (ICCSAI)*, 2025, doi: 10.1109/ICCSAI64074.2025.11064577.

[10] S. Agustri, E. Ermatita, and D. P. Rini, "Recommender systems for students' performance in higher educations: A review," in *Proc. 2023 Int. Conf. on Informatics, Multimedia, Cyber and Information Systems (ICIMCIS)*, 2023, doi: 10.1109/ICIMCIS60089.2023.10348622.